

# Energy and Mobility-Aware Routing for Disaster Response Drone Networks (FANETs)

## NS-3 Project Proposal: Implementing Fitness Function Routing

Mosharaf Hossain Apurbo

Department of Computer Science & Engineering

February 8, 2026

# 1. Introduction: The Scenario

## Real-Life Application: Disaster Response

- **Context:** A natural disaster destroys ground infrastructure.
- **Solution:** A swarm of Unmanned Aerial Vehicles (UAVs) creates a **Flying Ad-Hoc Network (FANET)**.
- **Goal:** Relay live video footage from the search area to a remote Base Station.

**The Constraint:** Standard protocols like AODV blindly select the shortest path. This drains specific "central" nodes and ignores drone speed, causing frequent link ruptures and mission failure.

## 2. Problem Statement

### The Flaw in Standard Routing (AODV):

- **Metric:** AODV uses *Hop Count* as the sole metric.
- **Decision:** "Path A (3 hops, but moving away at 50 m/s) is better than Path B (4 hops, hovering efficiently)."

### The Consequence

- 1 Critical nodes die prematurely (Energy Depletion).
- 2 Fast-moving nodes break links immediately (Route Instability).
- 3 High packet loss occurs during route reconstruction.

### 3. Project Roadmap: Core vs. Extension

I propose a robust implementation based on the fitness function methodology proposed by **Taha et al. [1]**, extended with mobility awareness to suit the FANET environment.

#### Phase 1: The Core Target (Reproduction)

##### Fitness Function AODV (FF-AODV)

- Replace "Shortest Path" logic with a multi-metric "**Fitness Function**".
- **Logic:** Calculate a score ( $F$ ) combining **Residual Energy** and **Distance**.
- *Status:* Base Implementation (Taha et al.).

#### Phase 2: Custom Modification (Mobility-Aware)

##### Velocity-Aware Optimization

- **Problem:** Taha et al. do not account for node speed.
- **Solution:** Extend the fitness function to penalize high-velocity nodes.

## 4. Methodology: Fitness Function (The Math)

Based on the approach by **Taha et al. [1]**, we define a "Fitness Score" ( $F$ ) for every route option.

**The Base Formula (Phase 1):** We aim to maximize  $F$ :

$$F = \alpha \times \frac{E_{residual}}{E_{initial}} + \beta \times \frac{1}{HopCount}$$

- $\alpha$  (Alpha): Weight for Energy (e.g., 0.6).
- $\beta$  (Beta): Weight for Distance (e.g., 0.4).

**The Algorithm Change:** In standard AODV, we update a route if  $Hops_{new} < Hops_{old}$ .

In **FF-AODV**, we update a route if:

$$F_{new} > F_{old}$$

## 5. Methodology: Custom Modification (Velocity)

For Phase 2, we introduce a **Normalized Velocity Factor** to handle the high-speed nature of FANETs.

### Velocity-Aware Fitness Function

We add a third parameter,  $\gamma$  (Gamma), to score stability:

$$F_{new} = F_{base} + \gamma \times \left( \frac{V_{max} - V_{current}}{V_{max}} \right)$$

*\*Note: If  $V_{current} = 0$  (Hovering), the score is maximized (1.0).*

### Integration Logic:

- ➊ **Step 1:** RREQ arrives at a node.
- ➋ **Step 2:** Node checks its own current speed ( $V_{current}$ ) vs Max Speed ( $V_{max}$ ).
- ➌ **Step 3:** Calculate Stability Score. (Faster nodes get lower scores).
- ➍ **Step 4:** Routing table updates only if the new path offers the best balance of Energy, Distance, **AND** Stability.

## 6. Implementation Details (ns-3)

We will modify the C++ source code in `src/aodv`.

### Step 1: Energy & Mobility Awareness

- Install `BasicEnergySource` on nodes.
- Access `MobilityModel::GetVelocity()` to fetch real-time speed data.

### Step 2: Modify Route Request (RREQ)

- **File:** `aodv-packet.h`
- Add field `m_pathFitness` and `m_velocity` to the header.
- As RREQ travels, intermediate nodes update these fields.

### Step 3: Decision Logic

- **File:** `aodv-routing-protocol.cc`
- In `RecvRequest()`:
- `double v =`  
    `m_ipv4->GetObject<MobilityModel>()->GetVelocity().GetNorm()`
- `Calculate F(Energy, Hops, v);`
- `if (newFitness > storedFitness) { Update Route; }`

## 7. Evaluation Strategy

We will compare **Standard AODV** vs. **FF-AODV (Method 2)** using 3 Key Metrics:

### 1. Network Lifetime

Time elapsed until the **first node dies**.

- *Hypothesis:* FF-AODV significantly outlasts Standard AODV by balancing the load.

### 2. Packet Delivery Ratio

Percentage of successful video packets.

- *Goal:* Maintain high reliability by avoiding fast-moving, unstable nodes.

### 3. Average Throughput

Rate of successful data delivery over the communication channel (Kbps).



## 8. Simulation Parameters

| Parameter                         | Value                                |
|-----------------------------------|--------------------------------------|
| Simulator                         | ns-3 (Latest Version)                |
| Area                              | $1000m \times 1000m$ (Disaster Zone) |
| Number of Nodes                   | 30 UAVs                              |
| Mobility Model                    | Gauss-Markov (Random Flight)         |
| Traffic Type                      | UDP (Video Stream)                   |
| Initial Energy                    | 100 Joules                           |
| Weights $(\alpha, \beta, \gamma)$ | 0.5, 0.3, 0.2                        |

Table: Proposed Simulation Setup

## 9. Expected Outcomes

- **Outcome 1:** A working "Fitness Function" inside the ns-3 AODV module (Phase 1).
- **Outcome 2:** Validated Results showing:
  - Increased Network Lifetime.
  - Load balancing (traffic moves away from low-battery nodes).
- **Outcome 3 (Custom Mod):** Improved Packet Delivery Ratio in high-mobility scenarios (20 – 50 m/s) by avoiding unstable, high-velocity nodes.

# References



[1] A. Taha, R. Alsaqour, M. Uddin, M. Abdelhaq and T. Saba,  
"Energy Efficient Multipath Routing Protocol for Mobile Ad-Hoc Network Using  
the Fitness Function,"  
*IEEE Access*, vol. 5, pp. 10369-10381, 2017.



[2] ns-3 Consortium,  
"ns-3 Network Simulator,"  
<https://www.nsnam.org/>.



[3] H. Yang and Z. Liu,  
"An optimization routing protocol for FANETs,"  
*EURASIP Journal on Wireless Communications and Networking*, vol. 2019, no.  
120, 2019.  
(Referenced for FANET mobility challenges)